

BURLINGTON, WI, USAWMO: **722059**Lat: **42.690N**Lon: **88.304W**Elev: **779**StdP: **14.29**Time Zone: **-6.00 (NAC)**Period: **04-19**WBAN: **04866****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-1.8	-15.3	2.5	-6.0	-9.7	3.4	-0.6	25.7	27.5	23.5	22.1	5.4	280	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	89.5	73.8	86.2	71.7	83.5	70.8	76.7	84.4	74.7	82.4	73.0	80.4	9.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.3	131.7	80.9	72.4	123.6	79.1	70.5	115.6	76.9	40.6	84.4	38.7	82.6	37.1	80.5	99.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.1	17.1	-13.9	94.9	7.6	6.9	-19.3	99.8	-23.8	103.9	-28.0	107.7	-33.5	112.7		
			DB													
			WB	-14.3	81.2	7.5	5.6	-19.7	85.3	-24.1	88.6	-28.3	91.7	-33.8	95.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	20.8	22.5	34.3	45.8	57.3	66.6	70.1	68.1	61.7	49.7	36.7	26.1
	DBStd	19.86	12.08	11.86	11.26	9.01	8.84	6.88	5.71	5.44	8.08	9.27	10.40	10.79
	HDD50	3677	906	771	505	185	30	0	0	0	6	121	411	742
	HDD65	7168	1371	1191	953	579	269	60	18	31	157	484	849	1206
	CDD50	2501	0	0	18	59	258	500	622	562	358	110	13	1
	CDD65	516	0	0	1	3	32	110	175	127	59	9	0	0
	CDH74	5230	0	0	11	59	420	1086	1714	1191	647	101	1	0
	CDH80	1513	0	0	1	7	114	345	512	316	195	23	0	0
	WSAvg	6.9	8.5	8.2	7.9	8.8	7.2	5.9	4.8	4.3	5.1	6.8	7.6	7.9
	PrecAvg	35.1	1.9	1.7	2.3	3.3	4.1	4.6	3.3	3.7	3.1	2.6	2.5	2.0
Precipitation	PrecMax	43.6	4.0	4.5	4.6	6.7	10.5	8.3	7.7	11.6	6.7	6.7	5.2	4.5
	PrecMin	25.9	0.4	0.1	0.6	1.4	1.1	0.9	1.1	1.3	0.5	0.4	0.5	0.7
	PrecStd	4.8	0.9	1.1	1.2	1.4	2.2	2.2	1.6	2.2	1.7	1.7	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.3	59.2	74.9	80.8	88.0	92.6	93.4	90.4	90.5	83.0	69.4	56.4
		MCWB	76.3	53.1	60.9	61.0	70.5	73.9	77.1	72.7	72.0	71.1	58.2	52.4
	2%	DB	47.1	48.3	63.5	74.2	83.2	87.9	88.3	87.0	85.6	75.7	61.7	50.0
		MCWB	46.2	43.2	55.5	58.8	66.9	72.4	74.9	72.0	70.9	63.4	53.1	47.6
	5%	DB	41.0	41.3	57.1	68.2	79.1	83.9	84.7	83.3	81.4	70.6	56.7	44.6
		MCWB	38.1	37.2	50.2	56.5	64.7	70.2	72.2	71.1	67.1	60.7	50.7	42.3
	10%	DB	36.6	37.1	51.6	63.0	74.2	80.8	82.1	80.9	77.3	65.2	52.4	40.5
		MCWB	35.0	34.6	46.3	52.7	62.5	68.1	70.7	69.3	65.9	58.5	47.2	38.0
	0.4%	WB	72.5	53.2	62.2	65.9	73.9	78.2	81.0	77.5	75.3	73.1	59.8	54.7
		MCDB	72.5	56.5	72.1	75.0	82.4	86.9	87.9	85.8	85.3	81.5	66.8	56.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.3	44.0	57.4	61.6	69.6	75.1	77.1	75.0	72.6	67.5	54.9	46.9
		MCDB	46.7	47.8	63.6	69.4	78.8	83.6	84.0	82.2	81.3	72.1	60.2	48.9
	5%	WB	38.6	38.2	51.8	57.6	67.1	72.4	75.1	73.3	70.5	63.1	51.5	42.6
		MCDB	40.4	40.8	55.2	67.7	75.5	80.6	82.4	80.5	78.3	67.2	56.5	44.4
	10%	WB	35.0	35.1	45.4	53.4	64.3	70.2	73.2	71.4	68.2	58.9	47.5	38.0
		MCDB	36.4	37.3	50.1	62.2	71.3	77.7	80.2	77.9	75.3	64.6	52.1	40.0
Mean Daily Temperature Range	5% DB	MDBR	14.5	16.0	17.8	20.8	22.4	21.6	22.0	22.2	23.8	20.7	17.2	13.9
		MCDBR	17.2	20.5	26.4	31.2	29.7	25.2	24.0	25.0	28.1	28.7	24.1	16.8
		MCWBR	15.0	16.4	17.9	17.9	15.5	12.3	11.9	12.2	13.8	16.6	16.6	14.5
	5% WB	MCDBR	15.3	19.7	23.4	28.0	25.0	21.5	21.6	20.9	24.2	23.5	22.0	16.3
		MCWBR	13.7	16.3	17.4	17.7	14.9	12.3	12.7	11.8	12.9	16.5	16.6	14.8
Clear-Sky Solar Irradiance	taub		0.287	0.310	0.341	0.391	0.413	0.434	0.430	0.419	0.389	0.344	0.305	0.283
	taud		2.391	2.332	2.374	2.258	2.247	2.231	2.241	2.268	2.332	2.461	2.507	2.442
	Ebn at Noon		271	282	286	278	273	267	267	266	266	267	264	263
	Edh at Noon		25	31	34	41	43	44	43	40	35	27	22	22
All-Sky Solar Radiation	RadAvg		506	811	1151	1462	1693	1923	1944	1685	1380	888	566	409
	RadStd		70	94	109	122	128	132	94	104	115	103	51	39

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page